



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (06/07/2026 – 12/07/2026)

WEEK 8

PROJECT NAME	CHANGE DETECTION
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PROJECT DESCRIPTION :

This project aims to detect changes between satellite images captured at different times using Deep Learning and Computer Vision techniques.

GROUP MEMBER :

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GROUP ID :

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GROUP GUIDE :

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COLLEGE NAME :

GATI SHAKTI VISHWAVIDYALAYA

06/07/2026	• worked on report, CD, required documents, files for final thesis submission.
07/07/2026	• worked on report, CD, required documents, files for final thesis submission.
08/07/2026	• worked on report, CD, required documents, files for final thesis submission.
09/07/2026	• worked on report, CD, required documents, files for final thesis submission.
10/07/2026	• worked on report, CD, required documents, files for final thesis submission.
11/07/2026	Holiday (2nd Saturday)
12/07/2026	Holiday (Sunday)

Next Week Plan	Successful execution of project, presentation, report, final thesis.
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REFERENCE:

- https://docs.qgis.org/3.44/en/docs/user_manual/working_with_vector/editing_geometry_attributes.html
- <https://www.earthdata.nasa.gov/learn/data-recipes/detect-environmental-change-using-sar-data-qgis>

SCREEN SHOTS:

Several researchers have utilized Landsat satellite imagery to monitor urban growth and land-use changes over long periods. These studies employed image differencing and classification techniques to identify newly developed urban regions. The availability of historical Landsat data has made it one of the most widely used datasets for change detection applications.

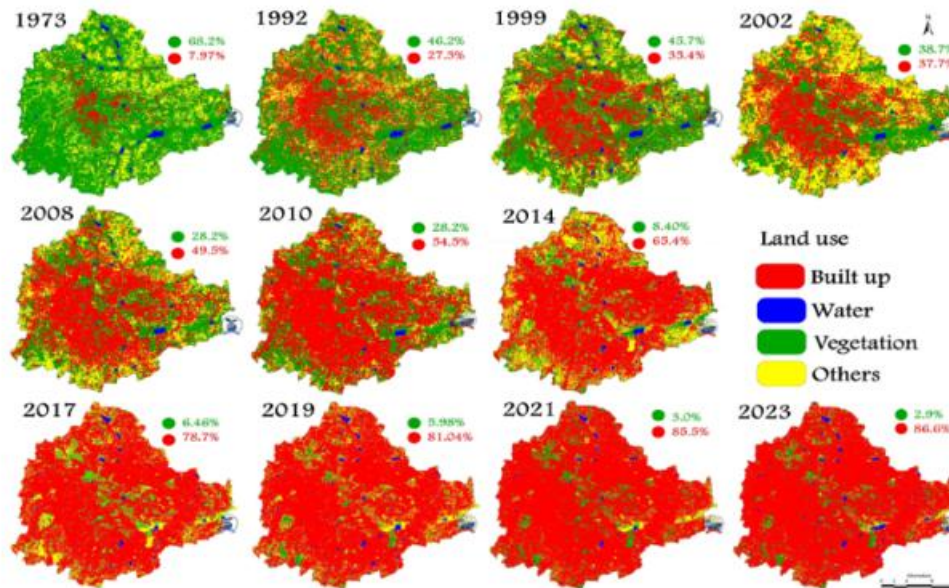


Figure 1.3: Landsat based change detection

1.6.2 Sentinel-2 Based Land Cover Change Analysis

With the introduction of Sentinel-2 satellites, researchers gained access to higher spatial and temporal resolution imagery. Various studies have used Sentinel-2 data to detect changes in vegetation cover, urban expansion, and infrastructure development. The multispectral

These were the objectives we worked toward:

1. To develop a web-based system for detecting changes using satellite imagery.
2. To acquire and process Sentinel-2 satellite data using Google Earth Engine.
3. To enable users to select an Area of Interest (AOI) for customized analysis.
4. To compare satellite images acquired at different time periods and identify changes.
5. To detect newly constructed structures and demolished areas within the selected region.
6. To monitor urban expansion and land-use changes using remote sensing techniques.
7. To provide an interactive visualization platform using ~~Streamlit~~ and Folium.
8. To display detected changes through maps and statistical summaries for easy interpretation.
9. To reduce manual effort involved in traditional change monitoring methods.
10. To support urban planning, infrastructure development, and environmental monitoring activities.